

HW Class12 Pt.2 (Population analysis) [Q13 Q14 BoxPlot]

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Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes. Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the medium value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `medium()` and `summary()` function that you can use to check your understanding.

```
library(dplyr)

# Read data
data <- read.table("rs8067378_ENSG00000172057.6.txt", header = TRUE)

# Find occurrence of label (genotype)
table(data$geno)
```

```
A/A A/G G/G
108 233 121
```

```
# corresponding median expression level for each genotype
median_expression_level <- data %>%
  group_by(geno) %>%
  summarize(median_value = median(exp, na.rm = TRUE))
median_expression_level
```

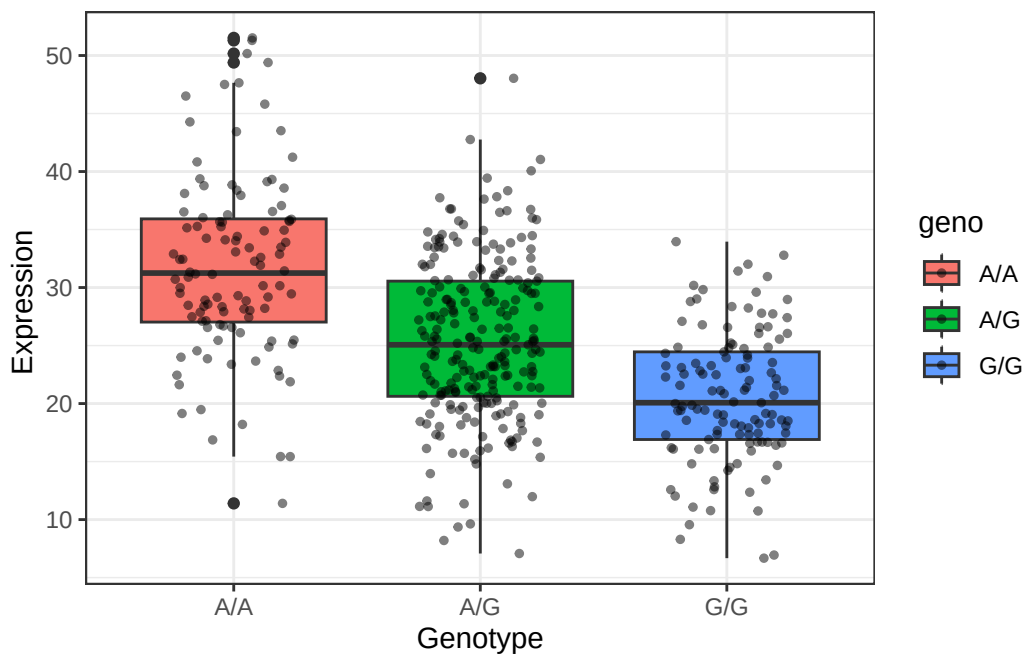
```
# A tibble: 3 x 2
  geno median_value
<chr>         <dbl>
```

1	A/A	31.2
2	A/G	25.1
3	G/G	20.1

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.

```
library(ggplot2)

# Create Boxplot
ggplot(data, aes(x = geno, y = exp, fill = geno)) +
  geom_boxplot() +
  geom_jitter(width = 0.25, alpha = 0.5, size = 1) +
  labs(
    x = "Genotype",
    y = "Expression "
  ) +
  theme_bw()
```



Ans: The boxplot shows that the relative expression value is greater for the A/A genotype compared to the G/G genotype. The SNP does seem to affect the expression of ORMDL3 as expression levels are dependent on genotype.